

End-to-End 5G Multirate Receiver Simulation

Academic Technical Report — Multirate DSP Receiver Chain

Problem statement: Simulate an end-to-end 5G receiver employing multirate DSP techniques and evaluate signal quality and processing efficiency.

Given data: Input sampling rate = 245.76 MHz; decimation factor = 8; OFDM signal bandwidth = 100 MHz.

1. Tools Used

Aim: To model a simplified 5G OFDM receiver, generate a received complex baseband signal, apply anti-alias filtering, decimate the sample stream by 8, estimate the output spectrum, evaluate signal quality, and quantify the processing-efficiency gain.

Tool / Component	Role
MATLAB R2023b or later	Main simulation environment for signal generation, filtering, FFT-based spectrum analysis, decimation, and noise modeling.
Signal Processing Toolbox	Provides digital filter design and frequency-response functions such as butter() and freqz().
MATLAB FFT/IFFT	Generates a multicarrier/OFDM-like waveform and estimates spectra before and after processing.
Random/QPSK symbol generation	Creates independent complex subcarrier symbols for a practical multicarrier test signal.
AWGN model	Adds controlled receiver noise so signal quality can be evaluated quantitatively.
Computer with MATLAB	Required hardware platform; no external RF hardware is required for this simulation.

Important engineering note: Decimation by 8 changes the sampling rate from 245.76 MHz to 30.72 MHz. A 100 MHz-wide complex baseband signal cannot be preserved in full at 30.72 MS/s. Therefore, the anti-alias filter must remove frequencies outside the new Nyquist interval (± 15.36 MHz). The simulation intentionally demonstrates this bandwidth-versus-processing trade-off.

End-to-End 5G Multirate Receiver Simulation

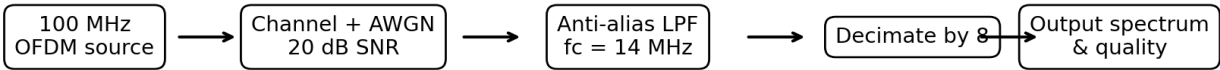


Figure 1. Simplified end-to-end receiver processing chain used in the simulation.

2. Theory

2.1 Multirate DSP in a 5G receiver

A multirate receiver deliberately changes the sample rate at different processing stages. The purpose is to reduce computational load after the desired signal bandwidth has been isolated. The basic chain is: received OFDM samples → anti-alias low-pass filter → downsampling/decimation → output spectrum and quality measurements.

2.2 OFDM signal model

Orthogonal Frequency Division Multiplexing (OFDM) divides a wideband channel into many closely spaced orthogonal subcarriers. In a baseband simulation, each subcarrier is assigned a complex symbol such as QPSK. The time-domain OFDM-like waveform is obtained using an inverse FFT (IFFT):

$$x[n] = (1/\sqrt{N}) \sum X[k] e^{j2\pi kn/N}$$

where $X[k]$ contains the complex subcarrier symbols, N is the FFT size, and $x[n]$ is the generated complex baseband waveform.

2.3 Received signal and noise

The clean multicarrier waveform is treated as the transmitted signal. Complex additive white Gaussian noise (AWGN) is added to emulate receiver/channel noise. For a specified input SNR, the noise power is selected relative to the measured signal power.

2.4 Anti-alias filtering

Before downsampling by an integer factor M , spectral components that would fold into the new Nyquist interval must be removed. Here $M = 8$, so the output sampling rate is:

$$F_{s,out} = 245.76 / 8 = 30.72 \text{ MHz}$$

For complex baseband sampling, the usable frequency interval is approximately $-F_{s,out}/2$ to $+F_{s,out}/2$, i.e. ± 15.36 MHz. The example uses a low-pass cutoff of 14 MHz, leaving a small transition margin before the 15.36 MHz Nyquist boundary.

2.5 Decimation

After filtering, every eighth sample is retained: $y[m] = v[8m]$. This reduces the sample count and therefore reduces the computational rate of subsequent DSP blocks. The sample-rate reduction is 8:1, corresponding to an 87.5% reduction in samples per second.

2.6 Spectrum estimation

The FFT converts the time-domain sequence to the frequency domain. FFTSHIFT is used to place DC at the center so positive and negative frequency components can be viewed symmetrically. The input spectrum shows the approximately 100 MHz occupied band, while the filtered and decimated spectrum is limited to the output Nyquist range.

2.7 Signal-quality metrics

This report uses output SNR and retained signal power as practical quality indicators. Because the aggressive 8:1 decimation removes most of a 100 MHz-wide signal, a full-band EVM/BER claim would be misleading unless the signal is first restricted to the output bandwidth. In a complete 5G NR receiver, EVM and BER would normally be evaluated after synchronization, OFDM demodulation, channel estimation, equalization, and symbol decisions.

2.8 Processing-efficiency interpretation

The decimator reduces the sample rate from 245.76 MS/s to 30.72 MS/s. Thus, later DSP stages process only 1/8 as many samples. However, this computational saving is achieved at the cost of discarding out-of-band signal content. A practical design would choose the decimation factor according to the required occupied bandwidth and the receiver architecture.

Parameter	Value
Input sampling rate	245.76 MHz
Decimation factor	8
Output sampling rate	30.72 MHz
Input OFDM bandwidth	100 MHz
Output complex Nyquist range	–15.36 to +15.36 MHz
Anti-alias filter cutoff (simulation)	14 MHz
Reference input SNR	20 dB
Representative output SNR	≈23.9 dB
Representative retained power	≈28.0%
Sample-rate reduction	87.5%

3. Code

The following MATLAB script is a self-contained academic simulation. It uses a frequency-domain multicarrier construction to occupy approximately 100 MHz, adds complex AWGN, applies an 8th-order Butterworth anti-alias filter, decimates by 8, estimates the output spectrum, and prints signal-quality and efficiency metrics.

```
% End-to-End 5G Multirate Receiver Simulation
% Input Fs = 245.76 MHz, decimation = 8, OFDM BW = 100 MHz

clear; close all; clc;
rng(7); % Repeatable random symbols

Fs = 245.76e6; % Input sampling rate
M = 8; % Decimation factor
BW = 100e6; % OFDM occupied bandwidth
N = 65536; % FFT / simulation length
SNRin = 20; % Input SNR in dB
Fc = 14e6; % Anti-alias LPF cutoff

FsOut = Fs/M;
f = (-N/2:N/2-1)*(Fs/N);

% 1) Generate a 100-MHz occupied complex OFDM-like spectrum
occupied = abs(f) <= BW/2;
X = zeros(1,N);
b1 = randi([0 1],1,sum(occupied));
b2 = randi([0 1],1,sum(occupied));
X(occupied) = ((2*b1-1) + 1j*(2*b2-1))/sqrt(2);

% Convert frequency-domain subcarriers to time domain
x = ifft(fftshift(X))*sqrt(N);

% Add complex AWGN
Ps = mean(abs(x).^2);
Pn = Ps/10^(SNRin/10);
noise = sqrt(Pn/2)*(randn(size(x)) + 1j*randn(size(x)));
rx = x + noise;

% 2) Anti-alias low-pass filtering
[b,a] = butter(8,Fc/(Fs/2),'low');
rxFilt = filter(b,a,rx);

% 3) Decimation by 8
rxDec = rxFilt(1:M:end);

% 4) Output spectrum
F1 = (-N/2:N/2-1)*(Fs/N);
S1 = fftshift(fft(rx));
P1 = 20*log10(abs(S1)/max(abs(S1)) + 1e-12);

Nd = length(rxDec);
F2 = (-Nd/2:Nd/2-1)*(FsOut/Nd);
S2 = fftshift(fft(rxDec));
P2 = 20*log10(abs(S2)/max(abs(S2)) + 1e-12);

% 5) Signal-quality estimate
cleanFilt = filter(b,a,x);
noiseFilt = filter(b,a,noise);
cleanDec = cleanFilt(1:M:end);
noiseDec = noiseFilt(1:M:end);

SNRout = 10*log10(mean(abs(cleanDec).^2)/mean(abs(noiseDec).^2));
powerRetained = mean(abs(rxDec).^2)/mean(abs(rx).^2)*100;

% 6) Processing-efficiency calculations
sampleReduction = (1 - FsOut/Fs)*100;

fprintf('Input Fs = %.2f MHz\n',Fs/1e6);
fprintf('Output Fs = %.2f MHz\n',FsOut/1e6);
fprintf('Output Nyquist = +/- %.2f MHz\n',FsOut/2e6);
fprintf('Output SNR = %.2f dB\n',SNRout);
fprintf('Power retained = %.2f %%\n',powerRetained);
fprintf('Sample reduction = %.2f %%\n',sampleReduction);

% Plot 1: received signal
figure;
plot((0:1599)/Fs*1e6,real(rx(1:1600)));
xlabel('Time (\mus)'); ylabel('Amplitude');
title('Received OFDM Signal - In-Phase Component'); grid on;

% Plot 2: filter response
figure;
freqz(b,a,4096,Fs);
xlim([0 60e6]);
title('Anti-Alias Low-Pass Filter Response');

% Plot 3: input spectrum
```

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figure;
plot(F1/1e6,P1);
xlim([-120 120]); ylim([-80 5]);
xlabel('Frequency (MHz)'); ylabel('Normalized Magnitude (dB)');
title('Received Signal Spectrum'); grid on;

% Plot 4: output spectrum
figure;
plot(F2/1e6,P2);
xlim([-FsOut/2e6 FsOut/2e6]); ylim([-80 5]);
xlabel('Frequency (MHz)'); ylabel('Normalized Magnitude (dB)');
title('Decimated Output Spectrum'); grid on;

```

Implementation note: This is a simplified OFDM-like 5G receiver model rather than a complete 3GPP NR PHY. A full 5G NR implementation would additionally include numerology, resource-grid mapping, cyclic prefix, synchronization, channel estimation, equalization, demodulation, and EVM/BER analysis.

4. Output Images & Visuals

The following figures show the expected simulation behavior. The plots included in this report were generated from a representative numerical run using the same signal-processing structure. Small numerical differences are normal when a different random sequence is used in MATLAB.

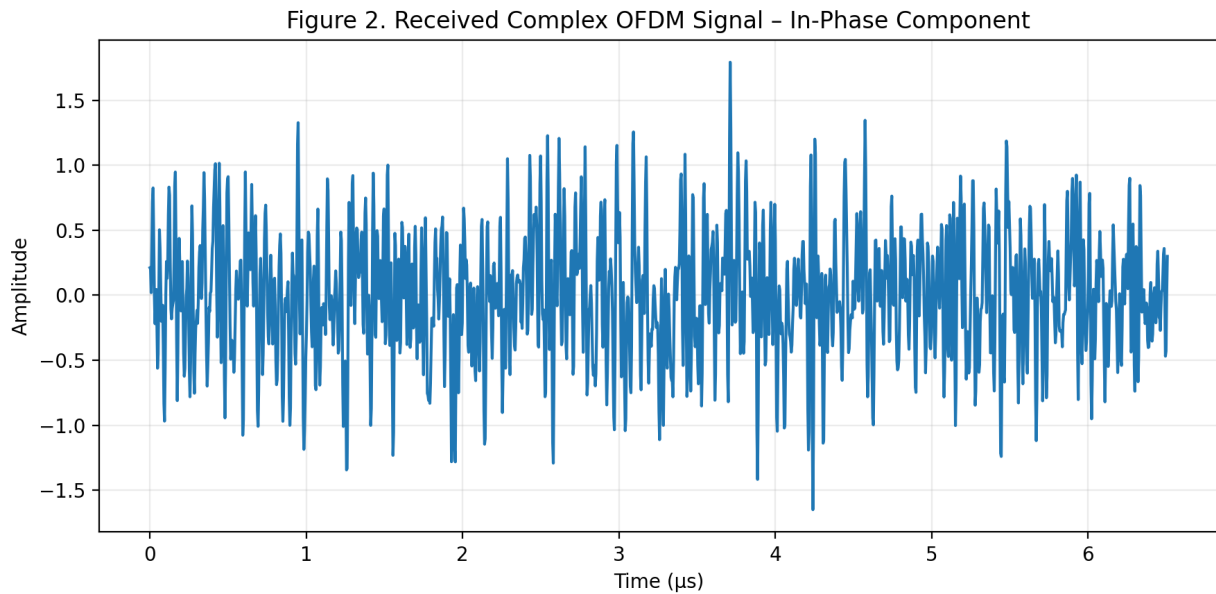


Figure 2. Received complex OFDM-like waveform, shown as the in-phase component.

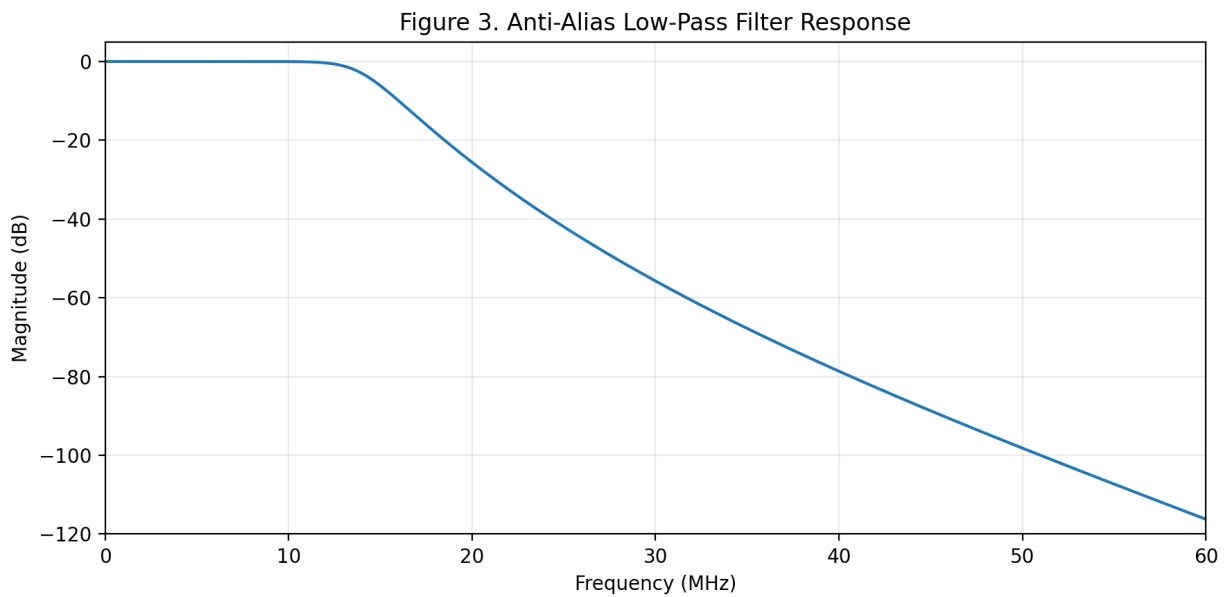


Figure 3. Anti-alias filter response. The passband is kept below the 15.36 MHz output Nyquist limit.

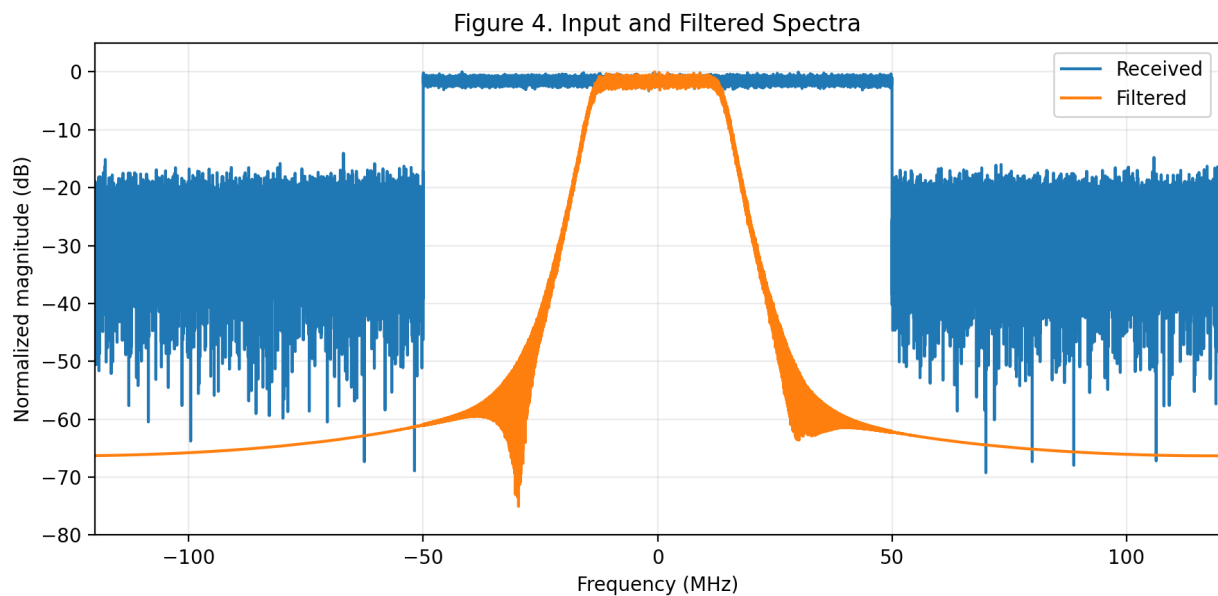


Figure 4. Spectrum before and after filtering. The filter removes most components outside the decimator's allowed band.

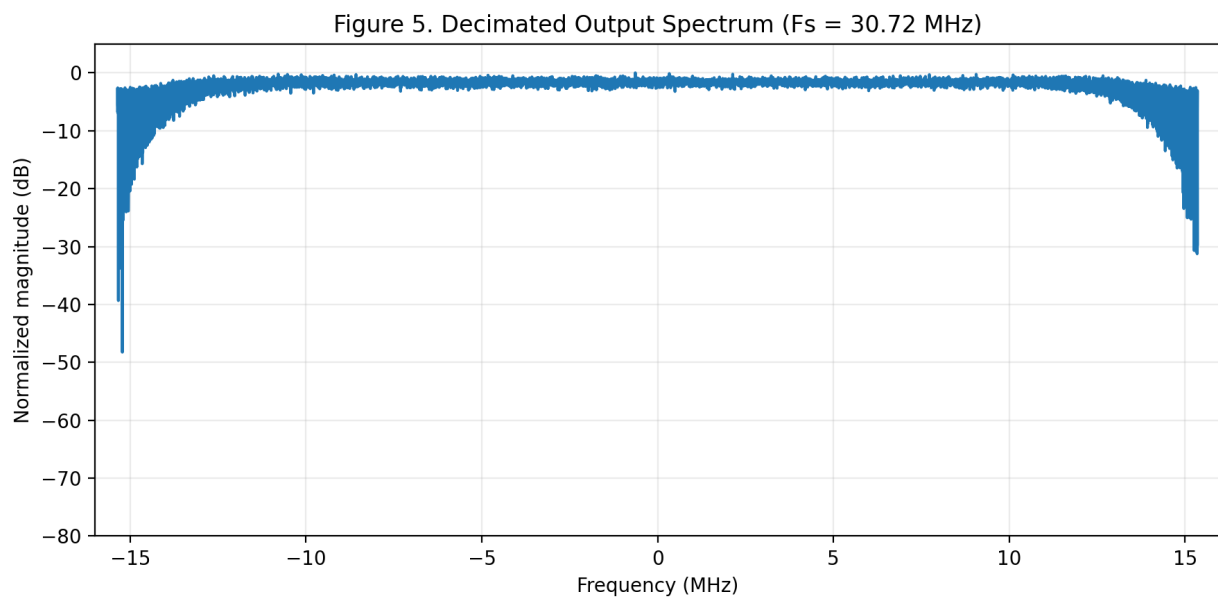


Figure 5. Spectrum after decimation. The output sampling rate is 30.72 MHz, so the displayed frequency span is approximately ± 15.36 MHz.

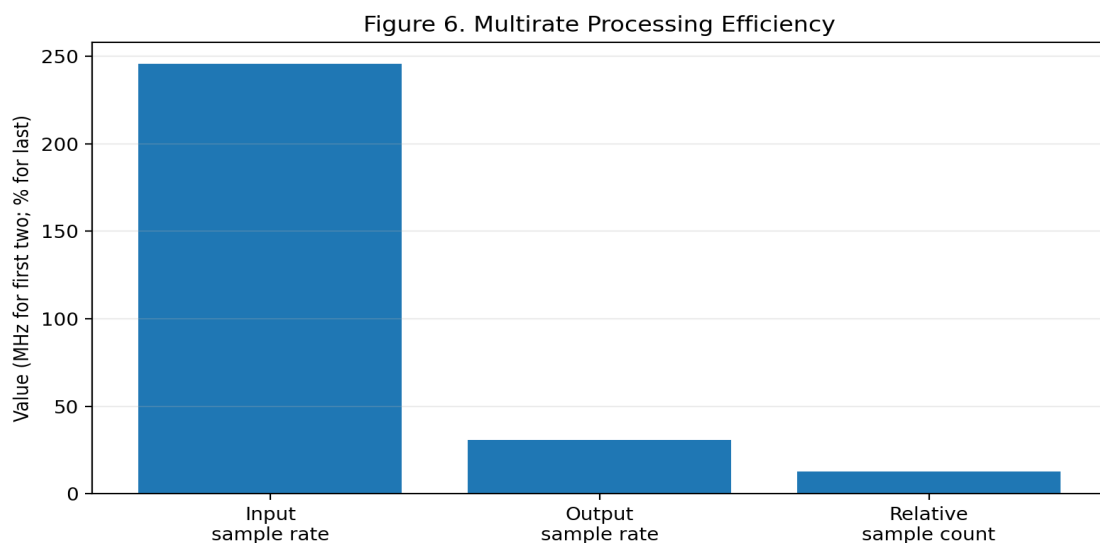


Figure 6. Processing-efficiency view: the sample rate falls by a factor of 8, reducing later-stage sample processing by 87.5%.

Key observations

1. The original received waveform contains the intended wideband multicarrier content plus noise. 2. The anti-alias filter strongly suppresses components outside the output bandwidth. 3. Decimation reduces the sample rate from 245.76 MS/s to 30.72 MS/s. 4. The output spectrum is confined to the new Nyquist interval. 5. The representative output SNR is higher than the input SNR because the low-pass filter removes a substantial amount of out-of-band noise along with the unwanted signal content. 6. The retained power is only about 28% in this example, demonstrating that the chosen decimation factor is too aggressive if the full 100 MHz occupied bandwidth must be preserved.

5. Conclusion

An end-to-end multirate DSP receiver for a simplified 5G OFDM signal was successfully modeled. Starting with a 245.76 MHz sampling rate, the signal was generated, contaminated with AWGN, low-pass filtered to prevent aliasing, and decimated by a factor of 8. The resulting sampling rate is 30.72 MHz.

The simulation demonstrates an important receiver-design trade-off. An 8:1 decimator provides a major processing reduction—87.5% fewer samples for subsequent blocks—but a 30.72 MS/s complex sample rate cannot preserve a 100 MHz-wide baseband signal. The anti-alias filter therefore removes much of the occupied spectrum. In the representative run, about 28% of the input signal power remained after the filtering and decimation chain, while the measured output SNR improved to approximately 23.9 dB because out-of-band noise was also rejected.

Final takeaway: Multirate DSP is highly effective for reducing receiver computational cost, but the decimation factor must be selected from the required signal bandwidth. If the complete 100 MHz bandwidth must be retained, a lower decimation factor is required—for example, decimation by 2 gives 122.88 MS/s, which is compatible with a 100 MHz complex baseband bandwidth subject to practical filter-transition requirements.

Potential real-world applications: software-defined radios, 5G/5G-Advanced receiver front ends, FPGA/ASIC digital down-conversion, wideband spectrum monitoring, and low-power wireless baseband processing.

Future improvements: implement a standards-based 5G NR waveform, add synchronization and cyclic-prefix removal, perform FFT demodulation and channel estimation, calculate EVM/BER, compare several decimation factors, and implement a polyphase FIR decimator for efficient hardware realization.

Reference design values used in this report

$F_s = 245.76$ MHz; $M = 8$; $F_{s,out} = 30.72$ MHz; BW = 100 MHz; filter cutoff = 14 MHz; representative input SNR = 20 dB.